Labour Mobility, Competing Auctions, and the Labor Share

A Comment on Mangin (2017)

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Abstract

Mangin (2017) shows that a competing-auctions labour market is constrained efficient for any admissible productivity distribution satisfying the model's regularity and entry conditions: the generalised Hosios condition holds endogenously without imposing parameter restrictions. I extend the model to a two-region economy in which workers may migrate at private cost. The central result is that the generalised Hosios condition extends fully to the migration margin — the decentralised allocation of workers across regions is socially optimal without any policy intervention. Under a natural labour-share ordering, migration also raises the aggregate labour share. A brief discussion of agglomeration and congestion externality identifies when and why efficiency at the migration margin can break down.

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I. Introduction

The Hosios (1990) condition characterises constrained efficiency in search models: the decentralised equilibrium is socially optimal when the relevant surplus share equals the corresponding matching elasticity. In Diamond–Mortensen–Pissarides models with Nash bargaining, this is often expressed as a restriction on workers' bargaining power and has no structural justification. Mangin (2017) shows that competing auctions change this fundamentally: when heterogeneous firms bid for workers in second-price auctions, the generalised Hosios condition holds *endogenously* for any admissible productivity distribution satisfying the model's regularity and entry conditions. Whether this property extends to a spatial economy with labour mobility is the question this paper addresses.

The question matters wherever large regional productivity gaps coexist with costly worker mobility. Indonesia provides a natural illustration: firms in Jakarta are substantially more productive than those in many eastern provinces, internal migration flows are prominent, yet their relationship to factor shares remains poorly understood (Badan Pusat Statistik 2023; Bryan and Morten 2019).

I extend Mangin's model to a two-region economy — a high-productivity Region H and a low-productivity Region L — in which workers face a private cost of migrating. Second-price auctions operate identically in both regions. The main result is that the generalised Hosios condition extends fully to the migration margin:

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workers' private migration decisions are socially optimal, and no policy correction is needed. The multi-region search literature (Lkhagvasuren 2012; Kawata, Nakajima, and Sato 2016) studies labour mobility under matching frictions but relies on Nash bargaining. In that setting, Kawata, Nakajima, and Sato (2016) show explicitly that the Hosios condition is not sufficient to restore efficiency at the migration margin. I also show that, under a natural ordering of regional labour shares, migration raises the aggregate labour share. A final section discusses agglomeration as an extension that breaks efficiency at the migration margin, motivating future work.

II. Baseline Model

I retain Mangin's notation. A measure L of risk-neutral workers and a continuum of potential firm entrants populate the economy. After paying entry cost C, each firm approaches one worker at random; the number approaching any worker is Poisson(θ), where $\theta \equiv N/L$ is market tightness and N denotes the measure of vacancies. Each firm draws match-specific productivity x from CDF G on $[x_0, \infty)$.

Workers post second-price auctions with reserve price w_R: in a *bilateral* meeting (one firm) the firm pays w_R; in a *multilateral* meeting (two or more firms) the highest-productivity firm wins and pays the second-highest productivity as the wage. Since firms bid their true valuation in second-price auctions, wages are determined competitively by the distribution of firm productivity.

The aggregate production function and expected output per match are:

$$f_G(\theta) = \int_{x_0}^{\infty} \theta e^{-\theta \bar{G}(x)} x g(x)\, dx, \qquad p_G(\theta) = f_G(\theta)/m(\theta), \tag{1}$$

where $\bar{G}(x) \equiv 1 - G(x)$ and $m(\theta) = 1 - e^{-\theta}$. The unique equilibrium has $w_R^* = z$ (the value of non-market activity) and market tightness satisfying the zero-profit condition (ZPC):

$$\int_{x_0}^{\infty} e^{-\theta \bar{G}(x)} \bar{G}(x)\, dx + e^{-\theta}(x_0 - z) = C, \tag{2}$$

which holds when the expected gain from entering equals the entry cost, ensuring no firm has an incentive to deviate.

Constrained efficiency in autarky. Workers' expected payoff simplifies to:

$$V(\theta) = f_G(\theta) - C\theta + z e^{-\theta}, \tag{3}$$

which is identical to the social planner's per-worker welfare $\Omega(\theta) \equiv f_G(\theta) + (1 - m(\theta))z - C\theta$. Because workers maximise the same objective as the planner — a direct consequence of the ZPC tying the expected wage to $p_G(\theta) - C/q(\theta)$ — the equilibrium θ^* is always constrained efficient. In this competing-auctions environment, the generalised Hosios logic appears in planner-FOC form as

$$f_G'(\theta^*) = C + z e^{-\theta^*}.$$

This condition holds endogenously, for any admissible G, without calibrating bargaining power.

III. Two-Region Extension

Setup. Region H has firm productivity distribution G_H on $[z, \infty)$ and Region L has G_L on $[z, \infty)$. Both regions share entry cost C and outside value z. Setting $x_{0i} = z$ in both regions eliminates matching rents ($\mu(\theta_i)(x_{0i} - z) = 0$), focusing the analysis on productivity rents and the tail index λ_i. Measures L_H and $L_L = L - L_H$ of workers reside in each region; workers in L may migrate to H at private cost $\kappa \geq 0$. Second-price auctions operate identically in both regions.

Productivity ordering. I assume Region H is more productive in the sense that it generates higher expected output: $f_{G_H}(\theta) > f_{G_L}(\theta)$ for all $\theta > 0$. I additionally assume

$$F_H^0(\theta) > F_L^0(\theta) \quad \text{for all relevant } \theta > 0, \tag{4}$$

where $F_i^0(\theta) \equiv \int_z^\infty e^{-\theta \bar{G}_i(x)} \bar{G}_i(x)\, dx$.[1] This ensures the Region H zero-profit condition binds at higher tightness ($\theta_H^* > \theta_L^*$) and that workers obtain higher equilibrium value in Region H: $V_H^* > V_L^*$.

Migration equilibrium. With moving costs distributed on $[0, \bar{\kappa}]$, workers with $\kappa_j < V_H^* - V_L^*$ migrate. An interior equilibrium exists for $\kappa^* \equiv V_H^* - V_L^* \in (0, \bar{\kappa})$, with equilibrium migrant mass M^*.

Lemma 1 (Regional independence)**.** *For each region $i \in \{H, L\}$, there exists a unique equilibrium tightness $\theta_i^* > 0$ satisfying the ZPC (2) with G_i in place of G, and $w_{Ri}^* = z$. Moreover, θ_i^* is independent of the worker measure L_i.*

Proof. See Appendix A. ■

The independence holds because free entry pins the *ratio* $\theta_i = N_i/L_i$: when L_i changes, vacancies adjust proportionally, leaving θ_i^* unchanged. The per-worker payoff is $V_i^* \equiv f_{G_i}(\theta_i^*) - C\theta_i^* + ze^{-\theta_i^*}$; conditions (4) and $f_{G_H} > f_{G_L}$ jointly ensure $V_H^* > V_L^*$.

IV. Main Result: The Hosios Condition Extends to Labour Mobility

Theorem 1 (Generalised Hosios condition with mobility)**.** *Suppose productivity distributions G_H and G_L are fixed (independent of the worker allocation). The two-region competing-auctions equilibrium is constrained efficient: the generalised Hosios condition holds within each region, and migration is socially optimal.*

Proof. Within-region efficiency follows from Mangin (2017) Theorem 1 applied to each region via Lemma 1. For the migration margin, since θ_i^* is independent of L_i, the social planner maximises

$$W(M) = (L_H^0 + M)\Omega_H(\theta_H^*) + (L_L^0 - M)\Omega_L(\theta_L^*) - \int_0^M \kappa(m)\, dm,$$

where $\kappa(m)$ is the m-th cheapest moving cost. The planner's first-order condition is $\Omega_H(\theta_H^*) - \Omega_L(\theta_L^*) = \kappa(M^P)$. Since $V_i^* = \Omega_i(\theta_i^*)$ in each region, this becomes $V_H^* - V_L^* = \kappa(M^P)$, which is identical to the decentralised migration condition. Hence $M^P = M^*$. ■

The result follows directly from Lemma 1. In standard spatial search models, migration creates a congestion externality: an arriving worker reduces tightness for those already in the destination region, but does not account for this when comparing payoffs. Under competing auctions with free entry, this externality is absent — tightness is pinned by the ZPC regardless of L_i — so the private payoff gain $V_H^* - V_L^*$ is also the social gain. The auction mechanism therefore satisfies the migration margin automatically, without any parameter restriction.

Remark 1. *This result stands in sharp contrast to Nash-bargaining search models. There, efficiency requires the Hosios condition within each region and a separate efficiency condition at the migration margin (Hosios 1990). Here, the ZPC automatically aligns the worker's private payoff with social welfare in each region, and Lemma 1 automatically aligns the private migration decision with the social optimum. The auction mechanism delivers efficiency at both margins without any parameter restriction.*

1. First-order stochastic dominance (FOSD) alone does not guarantee (4) because the integrand $ye^{-\theta y}$ is non-monotone in y. Condition (4) is therefore imposed directly as a productivity-ordering assumption. It is satisfied, for example, when Region H has a location advantage — a scale shift that raises average firm productivity — and is separate from the tail-thickness condition used later in the labour-share comparison.

V. Labour Share

In the Mangin (2017) model, the labour share is determined by two components: *productivity rents* (surplus the winning firm keeps because it is more productive than the runner-up) and *matching rents* (surplus bilateral-meeting firms capture due to monopoly power). Since we set $x_{0i} = z$, matching rents are zero, and the labour share is:

$$s_L(\theta_i^*; G_i) = 1 - \frac{\pi_{G_i}(\theta_i^*)}{p_{G_i}(\theta_i^*)}, \quad \text{for } i \in \{H, L\}$$

where π_{G_i} is expected productivity rents. For well-behaved productivity distributions, a key result from Mangin (2017) is that higher market tightness θ raises the labour share: more competition reduces the productivity rent gap between winner and runner-up.

For the regional comparison, I assume Region H has a thinner productivity tail. Under Mangin's Pareto parameterisation $\bar{G}_i(x) = (x_{\min}/x)^{1/\lambda_i}$, a *larger* λ_i means a fatter tail and higher productivity rents. So $\lambda_H < \lambda_L$ means Region H has less firm heterogeneity, lower productivity rents, and a higher labour share.[2] By Mangin's Proposition 4:

$$s_L(\theta_H^*; G_H) > s_L(\theta_L^*; G_L).$$

Migration and the aggregate labour share. The aggregate labour share is the output-weighted average across regions:

$$s_L^{\text{agg}} = \frac{L_H f_{G_H}(\theta_H^*) s_L(\theta_H^*; G_H) + L_L f_{G_L}(\theta_L^*) s_L(\theta_L^*; G_L)}{L_H f_{G_H}(\theta_H^*) + L_L f_{G_L}(\theta_L^*)}. \tag{5}$$

Since Region H has the higher labour share, shifting more workers to Region H raises the output weight on the high-labour-share term. Under the maintained condition that Region H has a higher labour share along the movement from the initial allocation $M = 0$ to the equilibrium allocation $M^* = M^P$, migration raises the aggregate labour share:

$$\text{More migration to Region } H \implies s_L^{\text{agg}} \text{ rises.}$$

Since Theorem 1 ensures migration reaches its efficient level under the auction mechanism, the resulting allocation is also associated with the aggregate labour share of the planner's allocation.

VI. Conclusion

I extend Mangin (2017)'s competing-auctions model to a two-region economy with labour mobility. The central result is that the generalised Hosios condition extends endogenously to the migration margin: the decentralised allocation of workers across regions coincides with the social planner's optimum, requiring no policy intervention. The mechanism is Lemma 1: because firms enter proportionally to workers under free entry, migration generates no congestion externality, and the private payoff gain from migrating equals the social welfare gain exactly. Under a natural ordering of regional labour shares, the efficient migration equilibrium is associated with a higher aggregate labour share, establishing a direct link between worker mobility and the distribution of output between labour and capital.

The framework points to several natural directions for future research. First, introducing agglomeration — whereby productivity in Region H increases with its workforce — would break efficiency at the migration margin, since migrants do not internalise the spillover they generate for the destination region; characterising the optimal corrective subsidy is a direct extension. Second, relaxing the free-entry assumption underlying Lemma 1 — as in capital-intensive or heavily regulated industries — may generate a congestion externality in the opposite direction: migrants dilute tightness without triggering compensating firm entry, potentially producing excessive migration and compressing the destination labour share. Both extensions are empirically relevant for economies such as Indonesia, where agglomeration forces and entry barriers coexist across regions.

2. This tail assumption is separate from the productivity ordering in (4). The two assumptions are compatible when, for instance, Region H has a location advantage (higher average productivity) while also having less dispersed firm productivity (thinner tail).

Finally, embedding the framework in the dynamic model of Mangin and Sedláček (2018) and introducing worker heterogeneity would allow analysis of cyclical labour-share dynamics and selection into migration within the competing-auctions setting.

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Appendix: Proof of Lemma 1

Existence. Define $F_i(\theta) \equiv F_i^0(\theta) - C$ where $F_i^0(\theta) \equiv \int_z^\infty e^{-\theta \bar{G}_i} \bar{G}_i \, dx$. At $\theta = 0$: $F_i(0) = \mathbb{E}_{G_i}[x] - z - C > 0$ by Mangin's Assumption 2. As $\theta \to \infty$: $F_i(\theta) \to -C < 0$. By the intermediate value theorem, a solution $\theta_i^* > 0$ exists.

Uniqueness. $\partial F_i^0 / \partial \theta = -\int_z^\infty e^{-\theta \bar{G}_i} \bar{G}_i^2 \, dx < 0$, so F_i is strictly decreasing and θ_i^* is unique.

Independence of L_i. The ZPC involves only the ratio $\theta_i = N_i / L_i$. Under free entry, vacancies N_i scale proportionally with L_i, so θ_i^* is determined solely by the ZPC and is independent of L_i.

Optimal reserve wage $w_{Ri}^* = z$. Workers maximise $V_i = f_{G_i}(\theta_i^*(w_{Ri})) - C\theta_i^* + ze^{-\theta_i^*}$ over w_{Ri}. The first-order condition requires $f'_{G_i}(\theta_i^*) = C + ze^{-\theta_i^*}$. The ZPC implies $f'_{G_i}(\theta_i^*) = C + w_{Ri} e^{-\theta_i^*}$ (Mangin 2017, Appendix A.1). Equating these gives $w_{Ri}^* = z$, confirming workers set their reserve price equal to the outside option. ■